Javier Castro

Student number: 263043

Artificial Intelligence & Computer Science

11th of April 2025

Word Count: 8907

Supervisor: Dmitrijs Dmitrenko

School of Engineering, University of Sussex

User interfaces for behaviour change towards sustainable mobility

“This report is submitted as part requirement for the degree of artificial Intelligence & Computer Science at the University of Sussex. It is the product of my own labour except where indicated in the text. The report may be freely copied and distributed provided the source is acknowledged. I hereby give permission for a copy of this report to be loaned out to students in future years”

Sustainable mobility in Madrid is the project on which this study is developing a prototype mobile application in a user-centered way integrating different means of transport↩. Spaces provided by: Analysis of existing mobility solutions—active transport (BiciMad, Lime), public transportation (Metro, buses, Cercanías, trams), high-occupancy vehicles (Wible, Uber, BlaBlaCar), and the role of private cars.

Madrid's campaign for sustainable mobility complements the government's aims to tackle emissions and congestion. There are also eco-friendly options, such as shared bikes like BiciMad and Lime, but they run into challenges in terms of regulation and use. Public transport is still fairly utilized, but private vehicles are better for convenience. Public system integration supported the transport pass and the Madrid Mobility 360 app. However, there are limitations to user awareness (e.g., users do not know where to get information about, e.g., integrated tickets) and system efficiency (in terms of multimodal transport interchange, for example, integration due to the limitations of the public system) which have been studied in the research paper.

A major survey carried out as part of the project showed that most Madrid residents use public transport or private vehicles, while shared mobility is underutilized due to worries over cost, accessibility, and safety. Although sustainability sounds great in theory, on a day-to-day basis, it is rarely used when participating in the commute.

To solve this, the project created an integrated mobility app (inspired by Moovit), designed with Figma and Anima, and developed it in React. It offers a multimodal route planner, carbon footprint tracker, real-time updates, customizable subscription models, and a community hub as key components. The feedback from user testing helped improve clarity, navigation, and usability, leading to a more accessible and user-friendly design.

Ultimately, this project is a triangulation of technical development, UX design, and user research that resulted in a prototype for more sustainable urban mobility at scale. The app provides citizens with a digital tool to help them choose greener travel options to support Madrid's transition to a sustainable transport future.

1. Introduction 5
2. Active mobility 7
    2.1 Bicimad 7
    2.2 Lime 8
3. Public transport 9
    3.1 Metro 9
    3.2 City and Intercity Buses 10
    3.3 Cercanías Renfe 10
    3.4 Metro Ligero (Trams) 10
    3.5 Transport Pass & Madrid Mobility 360 11
    3.6 Moovit 12
4. High Occupancy Mobility 14
    4.1 Wible 14
    4.2 Uber 15
    4.3 BlaBlaCar 16
5. Private Vehicle 18
6. Survays 20
7. App 22
    7.1 App Design 23
    7.2 User tester 25
    7.3 Source Code 27
8. Conclusion 28
9. Appendix 20
10. References 31

1. Introduction

In this project, we will study how Madrid brought about sustainable mobility solutions and create a prototype of an application in order to generate a more ecological transportation solution. Amid all this, cities across the world are placing a premium on more efficient, low-emission transport options, as fears about urban congestion and planetary sustainability grow. This project surveys the literature around existing mobility services (e.g., active mobility, public transport, and high-occupancy vehicle (HOV)) to extract best practices that can further inform the articulation of a new, user-friendly mobility app inspired by Moovit.

Sustainable urban mobility is critical not only for reducing carbon emissions, but also for making transport networks more efficient. The project aims to support this endeavour by analysing the workings of essential transport apps and gathering user feedback through surveys to develop a customized mobility service that meets the needs of commuters while also fostering environmentalism in journeying. The results will be used to create a state-of-art application system for mobility solutions.

This report is organised as follows: Chapter 2 presents an overview of active mobility solutions implemented in Madrid, namely bike-sharing services like BiciMAD, and scooter-sharing solutions like Lime. Chapter 3 explores the role of public transportation, examining Madrid's metro, city and intercity buses, Cercanías Renfe and tram services. Ride-sharing services such as BlaBlaCar and WiBLE and Uber-type car-sharing solutions — are discussed in Chapter 4. Lastly, in Chapter 5, an analysis of user feedback is presented, emphasising the main findings from surveys and how these will impact the design of the proposed mobility app.

Through analysis of what is already available and directly asking users what they want, this project presents an avenue for the development of sustainable mobility in Madrid and facilitation of the transition towards these types of habits.

The Spanish government defines sustainable mobility as understanding transport as a universal right, adapted to the needs of people, emphasizing active mobility and public transport as the two great alternatives for citizens. This policy seeks to promote well-being by implementing more active, accessible, and efficient mobility that has as its main objective to reduce emissions, improve air quality, and decrease the accident rate.

There are several types of mobility:

- **Active Mobility**: This category includes all journeys made on foot or by bicycle, including the mobility of people with disabilities or reduced mobility. It is located at the top of the so-called inverted pyramid of urban mobility (see Figure 1).
- **Collective public transport**: this includes the use of buses, trains and trams, since thanks to their low emissions they are key to reducing the number of private vehicles. It is located below active mobility in the pyramid of urban mobility.
- **High-occupancy mobility**: any vehicle rental service or sharing a vehicle on the same journey.
- **Private vehicle**: This category is at the bottom of the pyramid and consists, as the name suggests, of any private vehicle owned by one person. Priority should be given to using vehicles that involve lower emissions and less occupation of public space.

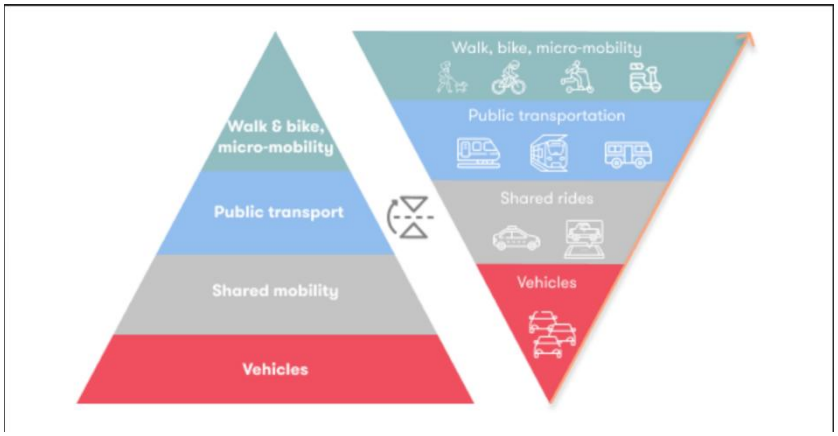


Figure 1: inverted pyramid of urban mobility

We’re well into the 21st century, where digital solutions are increasingly central to daily life. While not everyone has access to an electric scooter or bicycle, and some prefer alternatives to private cars, transportation apps are becoming indispensable. For those seeking comfortable, sustainable mobility options, we’ll explore the most important and widely used transportation apps available today.

2. Active Mobility

2.1 Bicimad

Bicimad is a bicycle rental service aimed at all citizens and visitors to the city of Madrid. It is an alternative form of transport that contributes to more sustainable mobility and encourages more balanced and healthy transport habits. This service was launched on October 28, 2022, and has been operating uninterruptedly 24 hours a day, 365 days a year since then.

The application works as follows: when you enter you will see the map of parking lots where the bicycles are located, these stations have a color code that informs of their status; the green color indicates low occupancy, the blue color indicates medium occupancy, the red color indicates high occupancy, and finally the gray color indicates that that station is not operational. Its use is very simple, once registered in the application you can go to any BiciMad station and take a bicycle, you can ride around the city for a certain time, and you will be charged according to the time of use. Once you have finished your trip, you must leave the bike at one of the stations. The bike must be properly attached to the base for the rental to end.

Beyond being a convenient transport option, BiciMad has demonstrated significant environmental benefits. A study conducted by the Polytechnic University of Madrid analyzed the impact of the service by measuring the distances traveled on BiciMad bikes. Initially, researchers estimated distances between stations using their geographic coordinates, but this proved inaccurate for urban travel. A more precise approach was developed using actual trip durations and the average BiciMad speed, estimated at 12 km/h based on EMT and media reports. With this method, accurate trip distances were calculated and compared to the likelihood of those trips otherwise being taken by car or bus.

For instance, a 13.5-minute ride, covering approximately 2.7 km, would statistically replace 7% of bus trips and 68% of car trips. By aggregating individual trip data, researchers estimated that in 2023 alone, BiciMad users saved approximately 451.8 tons of oil, 1,095.7 tons of CO_2, and 2.4 tons of NOx. These reductions are particularly beneficial in areas of Madrid where pollution levels are already high. Consequently,

BiciMad not only reduces emissions but does so at critical times and locations, significantly improving the city's environmental and energy balance.

km	Walk	Bike	Bus	Car
<0.2	94%	5%	0%	1%
0.2-0.4	81%	11%	0%	7%
0.4-0.6	64%	19%	0%	17%
0.6-0.8	38%	19%	1%	40%
0.8-1	56%	21%	1%	21%
1.0-1.5	25%	19%	3%	53%
1.5-2.0	18%	17%	5%	60%
2-3	10%	14%	7%	68%
3-5	4%	9%	10%	77%
5-7	1%	6%	11%	81%
7-10	1%	4%	12%	82%
10-20	0%	2%	10%	87%
>20	1%	1%	13%	85%

2.2 Lime

Lime is the world's largest electric vehicle sharing company, offering a future where transportation is shared, affordable, and sustainable. Lime offers electric scooters, electric bicycles, and in some cities, electric motorcycles. Through its mobile app, users can quickly and easily rent these vehicles to get around the city.

All these vehicles are spread throughout the city and can be left in permitted public places. Once you have located a nearby vehicle that you want to use, you can reserve and unlock it from the app as follows: first you will have to tap the icon of the scooter, bicycle, or motorcycle you want on the map. Once you get to the vehicle you will have to open the app and scan the QR code located on the handlebars of the vehicle, another way to do this is simply by manually entering the vehicle code.

2Once you have finished your trip, you must park in the permitted areas (without obstructing sidewalks or pedestrian accesses) and to finish, from the app you will have to select the **End trip option**.

The costs are based on the following: Lime will charge you a fee to unlock the vehicle (which varies depending on the type of transport) and then a fee per minute of use. For example, unlocking a scooter can cost between 1 and 2 euros, and the usage fee usually ranges between 0.15 and 0.25 euros per minute.

Lime focuses on sustainable mobility and its electric vehicles are designed to reduce C02 emissions and facilitate the reduction of traffic in large cities. In addition to all this, Lime also collaborates with local authorities to promote policies that favour the use of sustainable means of transport.

The micro-mobility company Lime has substantially impacted the city of Madrid. Through its dedicated Argentine environmentalists, it has promoted new, greener, and more sustainable alternatives to urban mobility. Below are Lime's main effects in Madrid:

Reduction in CO2 emissions and fewer car trips

During its first year of operation in Madrid, Lime users completed more than 1.5 million trips, thus avoiding around 80 tons of CO2. In turn, car trips avoided reached 650,000 and saved 175 tons of CO2.

Boost to sustainable mobility

Lime promoted shared electric vehicles; an absolute CO2 saving compared to private cars. According to Lime, a scooter trip consumes 75% less carbon than a car trip, and its entire fleet runs exclusively on renewable energy.

Regulatory and business challenges

As of October 2024, the Madrid City Council's Mobility Commission decided to revoke all licenses previously granted to electric scooter-sharing services, including Lime. The decision was based on the company's repeated disregard for parking regulations and the unsafe conditions caused to pedestrians. Lime has since appealed for an emergency injunction and continues expanding the service. Lime has played a significant role in reducing CO2 emissions and reducing the pressure on private car travel in Madrid, making it more environmentally friendly. However, the company still faces challenges operating in the city due to regulatory disputes.

3. Public Transport

Madrid has one of the most extensive and efficient public transport systems in the world, offering several options for getting around the Community of Madrid. The main forms of public transport include the **metro**, **buses**, **commuter trains** and **trams**. All of these modes of transport are managed by the **Madrid Transport Consortium** (CRTM).

3.1 Metro

The Madrid metro has one of the oldest and most extensive metro networks in Europe, with 13 main lines and 3 light rail lines covering the entire capital and some peripheral municipalities.

It generally operates from 6:00 to 1:30 in the morning, with train frequency varying depending on the time and line (at peak times it is more common for trains to run every 2 to 5 minutes).

Single ticket prices range from 1.50 to 2 euros (depending on the number of stations travelled). There are also 10-trip passes and monthly passes, which are more convenient for frequent users, but we will talk about this later.

3.2 City and Intercity Buses

The **EMT** or the **Municipal Transport Company** oversees managing all the buses in the city, while the intercity buses are managed by private companies under the control of the **CRTM**.

City buses cover almost the entire city of Madrid and connect parts of the city that are not served by the metro or the train. These buses operate from 6:00 to 23:30, but there are also some night lines, or night buses as they are known, which cover the city from 23:30 to 6:00. Single tickets cost 1.50 euros and can be bought directly when you get on the bus. It is also possible to use the transport pass or the multi-card for more frequent uses.

Intercity buses connect with the municipalities on the outskirts (metropolitan area). These buses generally operate throughout the day, and their frequency depends on the route and time of day.

3.3 Cercanías Renfe

This network connects the centre of Madrid with the surrounding municipalities and is key for longer journeys or reaching points of interest outside the centre.

It has 10 lines (C-1 to C-10) that run throughout the community of Madrid and connect to both the metro and the bus.

These trains generally run from 5:00 to 00:00 and frequencies vary depending on the line and time of day (every 5-10 minutes during peak hours). Prices vary depending on the zones. Fares are higher the further you travel from the city centre. It is also possible to use the transport pass for unlimited travel in specific zones.

3.4 3Metro Ligero (Trams)

The light rail system is a tram system that complements the metro network in the more remote areas of the city, especially in areas of recent urban development. There are 3 lines, ML1, ML2 and ML3, which cover different areas of the outskirts.

They follow the same zone fare system as the metro, and it is also possible to use the transport pass for these journeys.

Public Transport and Sustainability

Beyond providing mobility, Madrid's public transport system plays a crucial role in sustainability. A study by professors at Rey Juan Carlos University examined transport use patterns and their implications for sustainable mobility. Their research explored how factors like residential location, accessibility, and transport development influence commuting choices. The study found that while public transport is heavily relied upon, private vehicle use remains significant, revealing areas where public transport still struggles to compete.

One key conclusion was that neither the place of origin nor the destination alone determines transport choices—public transport is widely used in Madrid, but the continued preference for private vehicles suggests there is resistance to shifting away from car use. This may be due to congestion-related limitations affecting the speed and frequency of public transit. Researchers suggest that measures like restricting private vehicle use, expanding bike lanes, increasing pedestrianized zones, and enforcing stricter parking regulations for non-residents could encourage more sustainable commuting habits. Crucially, these policies could be implemented in a way that does not create social exclusion.

Madrid's extensive transport network is a backbone of the city's mobility, but further improvements in accessibility and efficiency could enhance its role in promoting sustainable urban travel.

3.5 Transport Pass & Madrid Mobility 360

A concept known as a transport pass has been discussed a lot. The transport pass is a rechargeable card that allows unlimited travel on all the means of transport within the

Community of Madrid that have been discussed. This pass is paid monthly, and the price varies according to the type of coverage. The types of passes are the following:

- **Youth pass**: this pass only covers people under 26 years of age and covers all zones for a fixed rate of 20 euros per month.
- **Normal Pass:** the price depends on the transport zones it covers and ranges from 54 euros in zone A to more than 131 euros for the most remote areas.
- **Senior Citizen's Pass:** This pass can only be used by people over 65 years of age and has lower rates than the rest.

The CRTM offers a mobile application called Madrid Mobility 360, which provides real-time information on the different transport options, fastest routes, fares and the status of vehicles. This application helps users to better plan their journeys.

3.6 Moovit

Moovit is an urban mobility app that aims to help users with their public transportation trips, combining different modes of transportation such as buses, subways, trams, trains, bicycles, and more. This app is used in several cities around the world and provides real-time information so that users can find the best routes and schedules. The app works as follows (see figure 2):

1. Route planning: Moovit allows users to check trips from a point of origin to the destination they want, choosing between several public transportation options. In order to achieve this, the app takes into account several possible routes, combining several forms of transportation to find the most optimal route.
    - Real-time route search: Moovit shows optimized routes and provides details such as the estimated arrival time, how many stops there will be, and the total trip time
    - Route alternatives: If the main route is very busy or congested, or there is a problem such as strikes or some type of service interruption, Moovit will offer alternative routes in real time.
2. Real-time information: One of the features that makes Moovit one of the best urban mobility apps is its ability to offer real-time information about the schedules and status of transportation. Thanks to this feature, users can see:
    - Real-time arrivals: Moovit shows the exact time left until the transportation you are waiting for arrives, whether it is a bus, train or subway
    - Updates on interruptions and delays: If there is any kind of problem with the transportation method, such as a delay, stoppages or detours, the app notifies users immediately.

- Alerts and notifications: Users can set up notifications to receive updates about their preferred line.

3. Interactive maps: Moovit provides interactive maps to facilitate navigation in cities, including:
    - Public transportation map: This feature shows the routes of all public transportation lines (buses, subways, trams, etc.) and allows you to see nearby stops.
    - Live tracking mode: Allows users to follow their trip in real time on a map, helping to visualize the progress of the journey and adjust a route if necessary.
4. Integration with other transportation methods: One of the advantages of Moovit is that it is not limited to just making routes considering public transportation, but the app can also be integrated with other mobility services, such as:
    - Bike and scooter services: Moovit allows you to see the accessibility of bikes and scooters in real time, making it easier to combine these modes with public transportation.
    - Taxis and rideshares: In some cities, Moovit also offers the option to integrate services such as Uber, Lyft, or other private transportation services to complete the trip.

Although Moovit is a very good way to organize a trip, it also has certain disadvantages:

- Although Moovit is very useful in many cities, its effectiveness depends on the coverage and quality of public transportation data in each area.
- Not all transports are always integrated: In some cities, the integration of certain private means of transport, such as taxis or car rentals, is not seamless

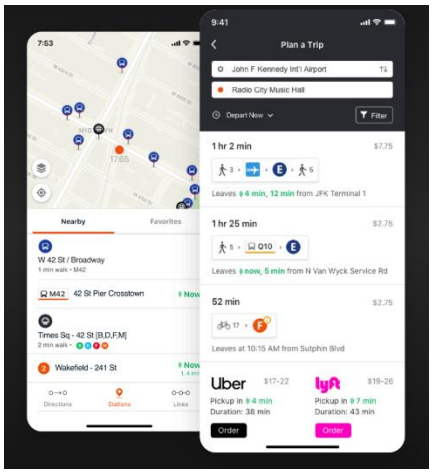


Figure 2: Moovit app

Moovit is a very useful app for planning trips using public transport, providing optimized routes, real-time information and details about different transport options, all in one app. It is ideal for both those who use public transport daily and for those who are simply visiting a new city and need guidance on how to get around.

Specific evidence would be needed to analyze Moovit's impact on reducing CO_2 emissions in Madrid; however, the initiatives and collaborations would reveal a commitment on the part of the company. Above all, the reasons for its use would positively influence commuter behavior. This is described in the Research in Transportation Economics publication, which highlights users' propensity to combine transportation, reducing the preference for private cars and the resulting emissions. Among the initiatives Moovit would join to strengthen its environmental responsibility

is Forever Green, which would use sports to achieve a positive impact; this impact would be achieved if "club members have agreed to work toward zero emissions in training and compensation." Consequently, Moovit has demonstrated community collaboration; However, data related explicitly to CO_2 emissions were unavailable, and it would correlate with the idea that more people are using sustainable services thanks to these initiatives.

4. High Occupancy Mobility

4.1 Wible

Wible is a carsharing app that offers a car rental service by the minute, hour or day and operates in Madrid. Wible was created by a collaboration between the companies KIA and Repsol, which makes it sustainable since all its vehicles are plug-in hybrids (PHEV), which helps reduce polluting emissions in the city. This app works as follows:

1. Download the app:
    - The app is available for both Android and iOS devices. Once downloaded, the user must register and follow some verification steps.
2. Location and selection of vehicles:
    - The app will show all available vehicles in real time on a map. Users can locate the closest one and easily reserve it.
3. Rates.
    - The most common and most affordable plan for short trips within the city is the per-minute plan, in which you pay depending on how many minutes you have used the vehicle.

    - The other plan that this company offers is the hourly and daily rate. This option offers cheaper rates if you are going to use the car for longer journeys.
4. Parking:
    - Users can park their vehicles only in designated parking areas, including coverage areas in Madrid or designated areas on the street.

Wible is a practical solution for those looking for more occasional mobility without compromising the environment, and without having to worry about fuel or parking in the city centre.

Wible is a carsharing service that offers zero-emission plug-in hybrid vehicles and operates in Madrid. As such, Wible impacts emissions reduction, which aligns with its primary objective of providing a sustainable mobility alternative. However, there are specific indicators for the exact amount of CO2 reduced by Wible in Madrid. This impact is reinforced through 100% green electricity consumption, making mobility a more sustainable practice than before. In addition to its effect on emissions reduction, Wible promotes sustainable mobility by offering an alternative to car ownership. As such, users can access Wible vehicles and do not need to use their private gasoline or diesel vehicles to get from one place to another. In this sense, Wible contributes to reducing traffic in the city, reducing polluting gas emissions.

Furthermore, the Wible carsharing service integrates with urban infrastructure based in municipalities surrounding Madrid, such as Pozuelo. As such, users can drive to the terminal in their private vehicle and switch to Wible there. These strategies demonstrate that Wible positively impacts Madrid's mobility landscape and reduces CO2 emissions.

4.2 Uber

Uber is one of the largest transport platforms in the world, connecting passengers with drivers through a mobile app. Users can request a trip from the desired location in a few steps from the app, and the company itself manages the rates, payments, and navigation in real time.

The operation of the app is very simple, once you have downloaded the application you only have to fill in some personal information and a payment method, either with a credit/debit card, with a digital account or even paying in cash at the end of the trip. Once the user has registered, they can request a trip by simply opening the app and selecting their current location, choosing the destination they want to reach, and the app will show them an estimate of the price and arrival time.

There are several types of services available.

- **UberX:** this is the most standard and cheapest option.
- **Uber Black:** this option allows you to travel in high-end vehicles, but its price goes up.

- **Uber Comfort:** these are cars that have more space and are used to have a more comfortable journey.
- **UberXL:** these vehicles are larger than the rest and can carry a group of up to 6 people.

- **Uber Green:** this option exists for those people who prefer to travel in electric or hybrid vehicles.

Uber has revolutionized urban transport in many cities, as it offers a very comfortable alternative to public transport or taxis. However, its environmental impact has been a topic of debate. A study conducted by Comillas University in Madrid examined Uber's use and its climate impact, highlighting concerns that the service increases pollution by introducing additional vehicles into circulation. On the other hand, research published by *Science Direct* in 2017 suggests otherwise.

A *Washington Post* article compiling such studies states: *"From hailing an Uber to using parks or public libraries, researchers have found that shared resources contribute to a decrease in individual energy consumption and help in the fight against climate change" (The Washington Post, 2017).* Additionally, Uber has been incorporating electric cars in various cities for years, a move that governments have encouraged to significantly reduce CO2 emissions in urban areas.

4.3 BlaBlaCar

BlaBlaCar is a carpooling application, or car sharing, that connects drivers who plan to take a trip by car with passengers who are looking for economical and efficient ways to travel. The app allows people going in the same direction to share a vehicle, saving costs and helping to reduce the carbon footprint.

The app works in a very similar way to the ones already mentioned (Wible or Uber), first you download the app and register, but in this case, users have to enter their origin and destination, as well as the date of their trip. BlaBlaCar will show them a list of drivers who will make the same trip that day, along with details such as price, departure time and seat availability.

If the user wants to offer the trip instead of searching for it, they will first have to publish the trip they are going to make in the app, indicating the origin, destination, date, departure time and the price per seat. They can also define the number of seats available, and detail whether they allow pets, whether they allow smoking in the car or even if they want the trip to be made in silence.

BlaBlaCar is a collaborative mobility solution that allows people to share trips economically and ecologically. This platform is ideal for medium and long-distance travel, and its shared economy model fosters trust, accessibility and sustainability in mobility.

BlaBlaCar: A Sustainable Carpooling Solution

BlaBlaCar is a carpooling platform that connects drivers with passengers traveling in the same direction, offering an economical and eco-friendly alternative to traditional transportation. By allowing people to share a vehicle, BlaBlaCar helps reduce travel costs and carbon emissions.

The platform operates similarly to services like Wible or Uber. Users register, input their origin and destination, and select a travel date. They are then matched with available drivers, with details on price, departure time, and seat availability. Drivers can also post their trips, specifying seat availability, pricing, and travel preferences (e.g., pet-friendly, smoking rules, or silent trips).

BlaBlaCar is particularly beneficial for medium- and long-distance travel, promoting accessibility, trust, and sustainability within mobility.

Environmental Impact of Carpooling

A BlaBlaCar study, *Zero Empty Seats*, highlights the platform's significant environmental benefits:

- **1.6 million tons of CO_2 emissions are reduced annually** through optimized vehicle use.
- **Average car occupancy on BlaBlaCar is 3.9 people per vehicle**, more than double Spain's national average of 1.7.
- **In Spain, BlaBlaCar users reduce CO_2 emissions by 42%** compared to traditional travel alternatives.
- **Over 25 million trips per quarter** occur via BlaBlaCar, with minimal added road congestion (+1.6% more cars but a 26% reduction in CO_2 emissions).

The 2018 study estimated that BlaBlaCar's impact was equivalent to removing all private cars from Barcelona or Valencia for a year. Future projections suggest that by 2023, **global CO_2 savings will reach 6.4 million tons annually**.

Changing Driving Habits and Road Safety

Carpooling is reshaping user behavior:

- **28% of passengers delay obtaining a driver's license**, opting for shared mobility.
- **22% of drivers reduce their speed while carpooling**, leading to lower emissions and improved road safety.

- **34% of Spanish users opt for more sustainable fuel alternatives** after joining the platform.
- **72% of users check their tire pressure more frequently**, improving fuel efficiency.

Spain stands out as one of the most efficient markets for carpooling, with 5 million users and an average trip distance of 285 km.

Technological Advances and Accessibility

Currently, **two-thirds of BlaBlaCar trips in Spain** lack a direct public transportation alternative. To bridge this gap, BlaBlaCar is implementing **artificial intelligence (AI) algorithms** to optimize routes and improve connectivity in underserved areas.

Key AI-driven improvements include:

- Reducing the **16 km average detour** for picking up/dropping off passengers.
- Increasing average car occupancy **from 2.2 passengers to 4** on routes between small towns.

BlaBlaCar's commitment to shared, multimodal, and efficient mobility aligns with regulatory frameworks like Andalusia's Climate Change Law, which incentivizes car sharing.

Jaime Rodríguez de Santiago, General Manager of BlaBlaCar Spain, Portugal, and Germany, emphasizes the importance of continued progress:
"We must work towards maximizing vehicle efficiency while encouraging regulatory changes that support sustainable mobility. The future of transport must be collaborative, multimodal, and environmentally responsible."

BlaBlaCar is proving that carpooling is more than just a convenient travel option—it's a transformative force in sustainable mobility. By reducing emissions, optimizing vehicle usage, and leveraging technology for smarter transport solutions, the platform is helping to shape a greener, more efficient future for urban and intercity travel.

5. Private Vehicle

Around 6 million people are living in Madrid, and according to the Madrid regional government, there are around 5.5 million private vehicles in circulation. According to an article released by the Ministry of Ecological Transition, and using data from 2021, the main actors that generate polluting gases in Spain are transport (with 29.9%). This negatively affects the environment in several ways:

- **Air Pollution**: Private vehicles emit polluting gases (C02, NOx and fine particles) that deteriorate air quality and can lead to respiratory problems and chronic diseases.
- **CO2 emissions**: Within the polluting emissions, the burning of fossil fuels (gasoline, diesel, gas) in combustion engines is what generates these greenhouse gases (GHG). These gases are a type of pollution that, in the long term, affects everyone and contributes to global warming and climate change.

Reducing the use of private vehicles and promoting sustainable alternatives, such as public transport, shared mobility and electric vehicles, are some of the strategies to mitigate these effects.

1	Mánchester
2	Moscú
3	París
4	Milán
5	Ámsterdam
6	Madrid
7	Londres
8	Copenague
9	Varsovia
10	Múnich
11	Berlín

Figure 3

Future Trends in Madrid's Mobility

By 2030, private vehicle use in Madrid is projected to decrease by 16%, marking a shift towards more sustainable urban mobility. According to Kantar's *Mobility Futures* study, which examines mobility in 31 global cities, sustainable transportation is expected to surpass private vehicle use for urban travel by this time. The report predicts that private car trips in major cities will decline from 51% to 46%, while public transport, cycling, and walking will make up 49% of all trips. The remaining 5% will consist of taxis, ridesharing, and other modes.

Among European cities, Manchester is expected to lead in sustainable mobility progress, followed by Moscow and Paris. Madrid ranks sixth in Europe and is the only Spanish city included in the study (See figure 3). Over the next decade, Madrid's transportation landscape is projected to change by 13.5%, affecting nearly 1 million residents. Specifically, public transport usage is expected to grow by 6%, cycling by 1%, and walking by 23%.

Measures to Support Sustainable Mobility

To ensure these changes take effect, Madrid is implementing measures such as restricting access for highly polluting vehicles in the city centre and expanding its public transport network. The city is also focusing on integrating green, accessible, and efficient transportation options.

Juan Luis Alonso, head of Mobility Futures for Spain at Kantar, highlights the challenges and opportunities of this transition:
"Madrid's population has doubled in the last 70 years, leading to an increase in private car usage and an expanded public transport system. While both modes dominate mobility today, the coming decade will bring significant changes. Madrid is becoming a national mobility laboratory. Balancing individual freedom with regulations for the

public good—such as improving air quality and reclaiming public space—will require time and effort, but by 2030, we will see a significantly transformed urban landscape."

Madrid's shift toward sustainable mobility represents a crucial step in improving air quality, reducing emissions, and enhancing the overall liability of the city.

This project aims to study different opportunities for sustainable transport and design a concept for an environmentally friendly mobility app inspired by Moovit. I am working on an app that would benefit from its integration into everyday routines.

To finalize this, I would investigate preexisting mobility apps and learn about their selling points, functionality, and user experience. Based on the best practices and innovative findings, I would select the components I would use in my prototype for optimal efficiency and usability.

I will also gather direct user feedback with surveys along with this technical analysis. They will be asked to provide feedback about the transport apps they use, what they find helpful, and which features they would like to see in an ideal mobility app. This is where surveys and interviews come into play and will come in handy to help understand what users want so the prototype satisfies real-life needs!

Through extensive research and user-driven design, this project seeks to develop a reliable, concise, sustainable mobility application that promotes accessibility and encourages users to opt for greener transport alternatives.

6. Surveys

Urban mobility is a key aspect of life; it shapes not only how residents move, but how they access and interact with their cities and its services. Here, we analyse the mobility habits of the Madrid inhabitants, providing qualitative information on which mode of transport citizens choose and the reasons behind their decisions, including which potential barriers they may face in changing their mode of transport or using one mode over another, etc. By identifying each of these influences, decision makers and transport planners can design better solutions to improve transport choices, reduce traffic, and promote green transport alternatives.

To this end, a large-scale survey based on diverse demographic and geographic backgrounds was conducted. This information makes it possible to get an understanding of how use is changing in the Madrid transport infrastructure, and where, if anywhere, the opportunities lie.

Madrid is one of the biggest and most dynamic European cities, which poses significant challenges for urban mobility. With population growth leading to increasing traffic on city roads, the pressure on the environment is also rising, and the need for more

sustainable and efficient means of urban transport is becoming clearer. The study at hand attempts to explore the mode choice of Madrid residents and to find out key factors that drive mode choice toward more sustainable mobility alternatives, given the pressing issues.

Demographic characteristics of respondents

The respondents spanned a range of demographics, providing a decent sense for how mobility breaks down by age and geography. There were 12 respondents from the 18-24 bracket, and 12 respondents between the age of 25-34. This demographic includes those who go into the labour force or into post-secondary education. Others are around the age of 35-44 (5 respondents), around the age of 45-54 (6 respondents) and over 55 (6 respondents). Categorize based on daily mobility behaviour at each stage in life.

In terms of geographical position, the majority of the people that did this survey (28) are living in core Madrid, while five are living in the periphery and eight in towns around. This difference stems from the disparity in mobility needs of urban and suburb dwellers. They are also more likely to live in city centres where there is better access to public transport (subway, bus, etc.), bikes and walking paths in general. Conversely, residents in the suburbs and outer towns are less dependent on public transit as public transit is less widespread or would require longer travel distances.

Mobility Reasons

Madrid's residents spend the majority of the calls they make within the city on work, leisure, shopping and education, according to the study. The largest proportion fall into work-related commutes, with 26 respondents stating that employment was their main reason for travel. Conversely, leisure and shopping become mobility demand factors for 18 respondents. For educational purposes, nine commute, especially as universities and schools contribute to daily transportation needs. One respondent, who travelled for another reason, confirms work, leisure, and education are still the primary drivers of urban mobility.

Transportation Modes

There are multiple options to get around Madrid, but public transit is the most common. The survey's results show that 24 respondents mainly use public transportation (like buses, commuter trains, and the metro). Private vehicles remain the second preferred mode of transport with 17 of its respondents relying on cars. Other forms of transport such as carpooling and cycling can also be used, however usage is still at a low level. Many issues prevent the use of these sustainable solutions: poor infrastructure, user convenience, and safety issues. The cost, journey time, and comfort were the most important parameters affecting the choice of a method of transportation, according to the respondents. It was recognised but being practical was more important than the environment.

Importance of Sustainability

While sustainability is playing a growing role in conversations surrounding urban mobility, the survey's results show that, for the most part, it is not yet front-of-mind for respondents when it comes to mode selection. By contrast, 10 respondents say it is "somewhat or less important," 14 are neutral on the matter, eight say it is "not very important" and three say it is "trivial."

These findings illustrate that, despite being aware and concerned about the environment, those residing in Madrid are not as affected by sustainability when it comes to their travel choices. It may also require initiatives encouraging to use sustainable transportation by promoting targeted incentives, improving infrastructure, and increasing awareness campaigns about the benefits of sustainable mobility solutions.

Data collection on use of shared mobility services

Even though the likes of Lime, BlaBlaCar Wible and others offering shared mobility services exist, they have always been limited to their use. Twelve respondents use these services infrequently; 26 said they never use them; one said he uses these services monthly; and only two respondents said they do use these services weekly.

The biggest barriers to using shared mobility services more often are high costs (5 answers), cumbersome registration (7 answers) and safety (15 respondents). These solutions are driven by prevailing issues that are a concern of modern life.

Car-Sharing Preferences

Among the 12 respondents who did use car-sharing programs, the biggest driving force was cost-effectiveness. Eight people decided because car-sharing always costs less than ownership of cars and taxis. Other reasons that were mentioned included flexible pickup and drop-off locations (4 responses) and environmental benefits (1 response).

Shared Mobility Improvements

Respondents identified several key improvements that would drive greater adoption of shared mobility services:

- Better integration with public transportation (14 votes)
- Ability to rent a car (10 replies)
- Better access to vehicles (9 mentions)
- Cheaper prices (8 replies)

These recommendations highlight the need for better pricing, usability, and seamless connectivity between different modes of transport in order to make shared mobility a more attractive proposition to people in Madrid.

Interest in Mobility Apps

89%) of those surveyed said they would be interested in an app that included all of Madrid's transportation options. Here are the most sought-after elements of such an app, according to the 34 respondents who supported the idea.

- Availability of shared cars in real time (34 responses)
- Cost comparison of different modes of transport(34answers)
- Finding multimodal routes (26 replies)

The implications of these results are that potentially, an integrated mobility platform can greatly enhance transportation options and help locals to pick a variety of modes that best suit their needs.

With a Madrid people mobility information of both private and public transportation, this study shows its open insight. Shared mobility services exist but their adoption is slowed by concerns on cost, access and safety.

However, significant interest in an integrated mobility app suggests there is an appetite among residents for innovative concepts that can enhance ease of use and connectivity in transport. Enhancing the integration of sustainable means of public transportation, addressing safety concerns, and investing in infrastructure perhaps would facilitate a shift towards more efficient and sustainable mobility practices in Madrid.

7. App

7.1 App Design

With increasing congestion in urban areas and climate change exacerbating the situation, sustainable mobility has never been more important. This combination must translate into real life, as we feel the need to adapt modern transportation systems to this goal, minimizing their impact on the environment while ensuring that public transportation is still as efficient and fully available. The app, which was designed on Figma, is a tool that helps users explore and adopt sustainable mobility options for a greener and more efficient urban ecosystem.

This is designed for urban commuters, eco-friends, and governments that want to change transportation habits for good. The solution is a one-stop shop to track carbon footprint, find eco-friendly transit options and—eco-responsibly—plan routes. Through a combination of these features presented to the end-user as a fluid user experience, the app creates a positive feedback loop that leads to green mobility.

More than just a navigation app, this shows users the ecological footprint of their transport choice. It links users to sustainable travel modes and encourages eco-friendly habits, motivating community movement toward a beautiful and liveable world. Together, we can all challenge the idea that sustainable mobility is simply a necessary

evil — and illustrate that it can be a lifestyle choice that makes all the difference. The citizens of the world must work together to reduce their carbon footprint, creating a future where the city gets less congested, the air quality improved, and transportation greener.

This app emerged from the necessity of a multimodal mobility tool in Madrid. With multiple ways to get around—public transport (bus, metro, and Cercanías), shared mobility (Wible, Lime, Bicimad), commuting with BlaBlaCar, and ride-hailing (Uber)—many users feel burdened with finding the best, most efficient, sustainable option. My vision was an app that integrates all these modes together into one, hybrid routing of multiple modes, and a monthly subscription-based payment system that is flexible based on their usage.

Starting with wireframes to outline the app's structure. I used Figma to create low-fidelity wireframes outlining the core components and user flow. Through iterative cycles and feedback, I distilled the ideas to high-fidelity prototypes. An emphasis was placed on clarity, consistency, and accessibility in the design of the UI elements.

Key design choices included:

- Green and blue colour palette to highlight sustainability and eco-consciousness.
- A typography that supports readability like a Sans-serif font for clarity and freshness.
- Easy navigation with a well-structured design to provide the user with an easy journey.
- Features and Functionality

Overflowing with essentials the app includes:

Multimodal Route Planner: provides the fastest and more sustainable routes by combining various transport modes into one journey (walking, using Bicimad and taking the metro in one trip).

Adaptable Subscription Model: Users can access a monthly payment plan with the transportation that they most frequently use (e.g., one plan with bus, metro, and Uber included, another covering Bicimad, Cercanías, and Wible).

Real-time transport updates: It provides live information on public transport schedules, shared mobility vehicle availability, and traffic conditions.

Rewards System: Incentives and achievements prompt users to select sustainable transport

Community Hub: user–initiated connections to sustainability initiatives and shared mobility programs in their local communities.

Fluid Navigation: An interface that operates seamlessly.

To make the app more appealing, elements like animated transitions and maps that update in real time were added.

The biggest challenge was uniting various transportation services to work as one, without sacrificing ease of use. To solve this issue, I worked on having a clear and intuitive layout making the journey planning of complex routes simple. Ensuring that the app is both responsive and accurate with real-time updates was another challenge. i.e. By using different transport provider APIs, I have managed to obtain real-time data and availabilities and journeys for the users.

And, building the model for subscriptions required significant insight into user preferences and commuter behaviour. This flexibility is key as the app offers personalized plans which can be tailored to suit your individual requirements, from getting on and off public transport, to walking and cycling.

A journey of merging creativity with usability the result is a clean, straightforward app that helps users overcome the obstacle of finding their way through Madrid's subpar transit system and encourages a switch to sustainable means of transportation.

As I move forward with this project, I intend to integrate more features such as real-time air quality data integration, AI route recommendation, and payment options to make it more accessible. Ultimately, they want to keep improving the app to make traveling sustainably as easy and appealing as possible for everyone.

This project has allowed me to dive deeper into UI/UX principles and the necessity of designing with the user in mind. You may have limited knowledge between training is complete, maybe in October 2023.

7.2 User tester

When we were prototyping app x, we ran user testing with five participants. This phase focused on testing the app's usability and flow with navigation high-fidelity prototype in Figma.

Select five users with various profiles (general and technical users). Every user used the prototype with a link and gave qualitative feedback in an interview afterwards. The sessions were 10 to 15 minutes long.

Users mentioned that while the overall navigation is smooth, there were times of confusion thanks to some unexplained icons and not being able to navigate back easily on some of the screens.

The design was well-received for its visual clarity. It was noted that a few components did not provide enough contrast, and that visual hierarchy could be further enhanced in places.

Several testers said that the purpose of the application wasn't clear to them when they first accessed it. This suggests an improved initial description.

It was observed that visual feedback was missed on buttons and interactions and more expected features like predictive search and indicator of actions should be added.

Proposed Changes

Given the feedback received we intend to introduce the following:

• Redesign the onboarding page with a clear and sharp explanation of the app.

• Enhanced contrast and visual hierarchy in sections that matter.

• Incorporation of micro-interactions and feedback screens for actions (e.g., errors, confirmations).

• Help elements like breadcrumbs or more direct buttons I would change out of platform navigation

Next Steps

After the changes have been implemented in Figma, a second round of testing will occur utilizing the same users (in an effort to validate the change) and new testers (to ensure clarity without any context).

In this 2nd stage of the user-centred design process, an updated version of the interactive prototype of the [Name of your app] app was tested. The previous version identified key areas to be improved–these include onboarding clarity, visual hierarchies, navigation and feedback of action. Armed with this knowledge, the design of the prototype underwent several notable changes and was tested again with a new group of users, excluding those who participated in the first session.

Test Objective

The aim of this second round was to check if the changes made have solved the initial issues found and to get a sense of how new users, without any knowledge of the app beforehand, would use it.

Methodology

The first batch of testers was completed and fully separate, nor do they know the users from the first wave, so got a quick and clear view of the application. Participants were provided with a link to the prototype in Figma and asked to carry out normal tasks such as navigating the app, identifying key features in different pages, effectively using the search engine, and perform simple actions. Then they filled out a brief interview or a form to describe their impressions.

Results and Analysis

All users responded that the navigation was simple and the design now sufficiently leads the user. The new “back” button received positive notices, and there was never any confusion when moving from screen to screen.

In fact, the introduction was rated as clear, direct, and useful, which solved one of the major shortcomings of the first version.

Three of four testers noted the redesign was easier to read due to higher contrast and stronger visual hierarchy, which helps users orient themselves inside the interface.

The addition of small visual features when performing an action, such as hover effects or visual confirmations, which were missing from the first version, was appreciated.

It was an enhancement, and it was well-received because it was valid. No new missing capabilities were found which are desperately needed.

Conclusions:

The changes made to the prototype were positively validated by the second round of user testing. The new testers reported that the main issues with the previous version were addressed, and the experience was fluid, clear, and satisfying. Navigation appeared accurate and users understood the app's function.

Future Recommendations:

These updates can be considered as minor improvements in a future functioning version in addition to the fact that the prototype is already in an advanced stage.

• Early users can optionally add on a quick, guided tour.

• Consider how easy it will be to access, or have shortcuts to, the app if it adds more sections in the future.

7.3 Source Code

The application described in this project was designed in Figma, a design platform popular among many web developers who use the tool to create user interfaces. To convert the design into functional code, it was necessary to use Anima, a plugin integrated into Figma that exports prototypes as React code. Although Anima was a quick and relatively easy way to generate the application's design structure, the generated code lacks organization, semantics, and good development practices, leading to additional optimizations and more readable and maintainable code. Due to this circumstance, it was necessary to refactor the code after exporting it through Anima. For this purpose, the files were reorganized, React and modern JavaScript programming best practices were added, and finally, the generated components were adapted to produce greater modularity and reusability. A metric called React Router was also added to manage the application's routing and allow smooth navigation between the different views that make up the interface. Several inaccuracies, such as component redundancy, poorly parameterized styles, and unclear structures, were discovered during this period. React Router's routing attribute was replaced with a more dynamic expression, so use Navigate. **To** is an attribute typically passed to the React Router component. When a link is clicked, the property is called. This route determines the exact location where the

link will be redirected. Using **Link** and **NavLink** is a way to implement declarative routing based on a JSX file rendered directly in the interface.

The **useNavigate** hook enables programmatic navigation from the same function or event. This means the redirect is triggered in response to a user action, such as a click, form validation, etc., or to more complex business logic. It provides greater control over the navigation pipeline and allows for a device when logical conditionals are needed before switching to a new route. While the **to** hook is appropriate when conditional logic isn't needed and with simple static bindings, **useNavigate** is much less restrictive, allowing for more customizable application pipelines. It's best used with dynamic logic and, above all, route validation. This is why it was implemented in this project; it improves the logical structure and shapes the navigation pipeline in response to user interactions. This project has demonstrated how effectively this application helps optimize the workflow from Figma to Anima and then to the React template. It has also emphasized the importance of subsequent project refactoring for professional maintenance. Using the useNavigate best practice for programmatic navigation, the application is now more robust, easier to scale, and maintain.

8. Conclusion

It has covered the complexities of psycho-social dynamics in urban mobility and commented on the implications of the human dimension for the future of urban transport. The project has analysed existing mobility services such as public transport, shared mobility platforms, and active transport alternatives, identifying the advantages and flaws of the current mobility ecosystem. A clear pattern is emerging through the integration of academic/gov types of technical research, user feedback, and real-world case studies. Currently, many solutions are available, but there is no coherent and unified platform.

The solution to these gaps is the design and prototyping of a mobility application, which is the subject of this report. The solution predicts optimal driving routes per user in real-time by correlating location-based data collected from users and aggregating it with ride-hailing carpark information from multiple systems across the city, drawing across user journeys from existing successful systems like Moovit to provide users easy, quick access to multimodal transport through one single platform. Udacity has trained dedicated data analysts who are proficient in organizing and processing data and equipped with their app to quickly plan their travel in an energy-efficient manner by tracking their carbon emissions and providing eco-friendly alternative routes.

Working across tools like a Figma canvas, the Anima extension (for exporting designs), and then refactoring code with React and **useNavigate** is a professional front-end. Beyond the technical aspects, user testing ensured the app was usable, adaptive, and appealing, making clear that design decisions closely matched commuter expectations.

According to the study, user interest in these new mobility and sustainability tools was quantitatively strong. The aggregate in-place surveys demonstrated significant potential for services, yet practical limitations, including safety, pricing models, and access, also existed. Overcoming this threshold will be critical to enabling a larger-scale uptake of shared and alternative mobility solutions in the city of Madrid.

In summary, this endeavour serves as a stepping stone toward an exciting mobility application and adds to the dialogue on sustainable urban transport. As cities like Madrid strive to cut carbon emissions and reimagine public space, digital tools like the one proposed here can act as an essential catalyst in that shift — empowering citizens to make greener, smarter journey decisions.

9. Appendix

Link to see the app in figma:
https://www.figma.com/design/lAObfYJ0OWVSOcGILlZNL5/Untitled?node-id=0-1&t=Vw2nRhXrDgT0uMKc-1

10. References

- (2020). *Wible App*. Wible. https://www.wible.es/como-funciona/

- (2023). *Sustainable mobility takes precedence over private vehicles in Madrid*. Gobierno de España. https://www.madrid360.es/la-movilidad-sostenible-se-impone-al-vehiculo-privado-en-madrid/

- (2020). *How has mobility changed in Madrid? One million fewer trips by public transport and cars as before*. El País. https://elpais.com/espana/madrid/2021-12-12/como-ha-cambiado-la-movilidad-en-madrid-un-millon-de-viajes-menos-al-dia-en-transporte-y-trafico-como-antes-de-la-pandemia.html

- (2024). *What is the 'fault' of cars (and transport) in CO2 emissions and climate change?* El Español. https://www.elespanol.com/motor/20241101/culpa-coches-transporte-emisiones-co2-cambio-climatico/897660554_0.html

- (2017). *CO2 emissions from cars: Facts and figures (infographic)*. European Parlament.https://www.europarl.europa.eu/topics/es/article/20190313STO31218/emisiones-de-co2-de-los-coches-hechos-y-cifras-infografia

- (2024). *Trams in Madrid*. Civitatis. https://www.disfrutamadrid.com/tren-cercanias

- (2024). *BiciMad*. MADRID CITY COUNCIL. https://www.bicimad.com/

- (2024). *Moovit app*. Moovit. https://moovit.com/es/

-(2019) University of comillas. *An analysis of Uber's key success factors*. https://repositorio.comillas.edu/rest/bitstreams/295728/retrieve

-(2024) Sustainable urban mobility and BiciMaD: An analysis using predictable models [Industrial Organization Specialty, Polytechnic University of Madrid]. Halcón Gibert. https://oa.upm.es/83903/1/TFG_GONZALO_HALCON_GIBERT.pdf

-(2014) Miguel, J. P. M., De Blas, C. S., & Barandalla, I. C. J. An empirical study on the use of public transport in the Community of Madrid as a key factor in sustainable mobility. https://doi.org/10.1016/j.cesjef.2013.12.001

-(2020) Private car use in Madrid will decrease by 16% in 10 years. https://www.kantar.com/es/inspiracion/movilidad/el-uso-del-coche-privado-en-madrid-disminuira-un-16-en-10-anos

-(2024) Car sharing reduces 1.6 million tons of CO2 per year. Herranz, S. B., & Herranz, S. B. Newsroom BlaBlaCar. https://newsroom.blablacar.es/noticias/el-uso-del-coche-compartido-reduce-1-6-millones-de-toneladas-de-co2-al-ano-3?utm_source=chatgpt.com#el_carpooling_ha_cambiado_los_habitos_de_conduccion_de_los_usuarios_